

SAFE OPERATING PROCEDURE DURING DRILLING & BLASTING OPERATIONS

01. Engage only authorised persons in the drilling operations.
02. Remove all the loose debris at the sink bottom, prepare hard ground & wash with water and carefully examine for misfires & sockets, before starting drilling operations.
03. Mark the centre line and drill periphery holes correctly to avoid any under & over excavation.
04. Use templates for proper direction and spacing of holes while drilling.
05. Use appropriate wooden pegs for plugging the drilled holes.
06. Wear personal protective equipment (goggles, ear plugs and hand gloves) when using compressed air for cleaning the drill holes.
07. Engage only minimum number of persons for charging of holes.
08. Remove all the drilling accessories from the bottom of the sink before lowering the explosives & detonators for charging the holes. Priming to be done only at the shaft bottom / face by the shot firer himself.
09. Disconnect power at the shaft bottom before charging is commenced.
10. Authorised shot firer himself shall charge the holes.
11. Do not use any metallic tools for charging the holes.
12. All the joints of the detonator leads will be insulated.
13. Care should be taken to protect the drainage bore hole from damage while blasting.
14. Circuit should be tested only with approved circuit tester.
15. Only approved type of exploder will be used for blasting purpose. The shots should be fired only from surface from a safe place.
16. Before connecting the leads to the exploder for blasting, ensure that all persons in the vicinity are cautioned & taken shelter.
17. No person will be allowed to the shaft bottom unless the fumes are cleared and ventilation fan is running.
18. Proper dressing of the sides & cleaning of scaffold will be made & sides and bottom of the scaffold will be washed thoroughly with water before starting mucking operations.
19. Proper examination will be made for misfires and if found it should be dealt as per Reg.No.177 of CMR 1957.

Responsibility: Drillers and shot firer

11.	Blasting	5	5	3	75
(a)	Storage of explosives	5	5	3	75
	*Note: Magazine approved by Chief Inspector of Explosives is provided. It is generally observed that no incident / accident take place in storage at our magazine. Action plan: 1) Standard guidelines for storage of explosives at magazine will becomplied. 2) Reserve stations will beprovided nearer to the working places Responsibility: 1) Asst. Manager, Magazine Incharge 2) Overman and Shotfirer. Expected date of Completion: Continuous & Plan existing.				
SI.No	Type of Hazard	C	E	P	Rank
(b)	Transport of explosives	5	5	3	75
	Attention Required: Damaged canisters, carrying both explosive and detonators in the same canister will beprohibited. Action plan: Proper canisters will beprovided with lock and key separately for explosives and detonators. Responsibility: Asst.Manager, Magazine Incharge, Shotfirer. Expected date of Completion: Continuous & Plan existing.				
(c)	Handling of explosives	5	5	3	75
	Attention Required: Pilferage of explosives, not returning the unused explosives. Action plan: Strict compliance of regulations and various guidelines given by DGMS on use and prevention of pilferage of explosives will beenforced. Responsibility: Assistant Manager , Overman, and Shotfirer. Expected date of Completion: Continuous & Plan existing.				
(d)	Use of explosives	5	5	3	75
	Attention Required: Improper pattern and charging of drill holes, not following blasting card system, in sufficient length of shot firing cable, not using of appropriate tools for charging and blasting, incorrect dealing of misfires may result in accidents. Action plan: Strict compliance of regulations and various guidelines given by DGMS on use of explosives will beenforced. Responsibility: Overman and Shotfirer. Expected date of Completion: Continuous & Plan existing.				
e)	Projectiles	1	5	3	15
	Attention Required: Projectiles can cause harm to nearby work persons. Various sources of projectiles are: flying fragments from blasting, Action plan: 1) All persons shall take appropriate shelter during blasting. Responsibility: 1) Site Incharge 2) Sirdar, Shotfirer, workmen Expected date of Completion: Continuous & Plan existing.				

13	Blasting	Action plan	responsibility
a)	Storage of explosives	i) Comply with Regulation and guidelines issued by DGMS and Chief Controller of Explosives.	Asst. Manager, Magazine Incharge
		ii) Provide Reserve stations nearer to the working places.	Overman and Shotfirer.
b)	Transport of explosives	Use proper canisters with lock and key, separately for explosives and detonators.	Asst.Manager, Magazine Incharge, SF.

c)	Handling of explosives	Strictly comply with Regulations guidelines issued by DGMS on use and prevention of pilferage of explosives.	Under Mgr. Overman, and SF
d)	Use of explosives	Strictly comply with Regulations guidelines issued by DGMS on use of explosives.	Overman and Shotfirer.

The relevant regulations on the above subject are Reg 183 to Reg 207 of CMR 2017

Stream crossing:

Reg 149 (6) of CMR 2017.

149(6) No working will be made in any mine vertically below-

(a) any part of any river, canal, lake, tank or other surface reservoir; or

(b) any spot lying within a horizontal distance of 15 meters from either bank of a river or canal or from the boundary of a lake, tank or other surface reservoir,

except with the previous permission in writing of the Chief Inspector and subject to such conditions as he may specify therein.

However, in addition to the above sufficient coal will be left in underground as per the recommendations of Subsidence Prediction study done by CIMFR

DETAILED ASSESSMENTS FOR MAJOR HAZARDS

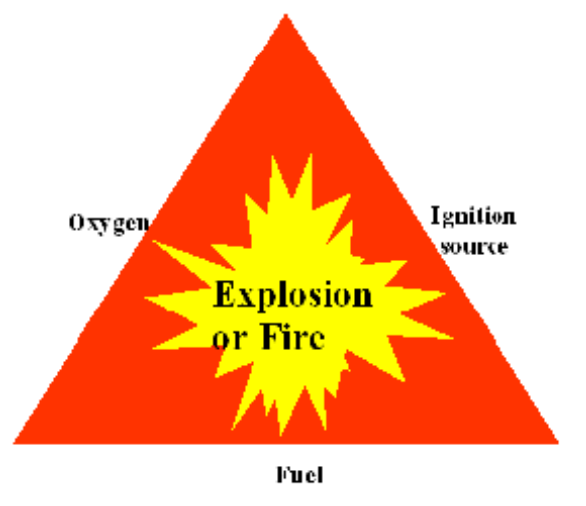
13.1 FIRE:

13.1.1 FIRE RISK ASSESSMENT

A fire risk assessment will help us to determine the chances of a fire occurring and the dangers from fire. The assessment method used is the same as for other hazards. Any fire in a mine is a devastating hazard. There are chances that any fire can cause loss of many valuable lives if the persons cannot escape in time. Even if persons can escape to safety, if the fire becomes uncontrollable the event may result in closure of the mine and huge loss to the company. Hence we have devoted more attention, a lot of time and space for a separate exercise on fire risk assessment. We have checked the existing fire safety arrangements in the mine and workplaces to satisfy the mine safety legislation. This check covered all the matters required by the Coal Mines Regulations, and conditions have remained unchanged, eg: numbers of people present, work activity etc.,.

13.1.2 The fire triangle:

For a fire to start three things are needed; namely a source of ignition; fuel; and oxygen. If any one of these is missing, a fire cannot start. Taking steps to avoid the three coming together will therefore reduce the chances of a fire occurring.



13.1.3 Fire Hazard: The following code will be adopted for fire control by minimizing sources of ignition at Jaganathpur B coalmine:

- a) Planning, installing and maintaining all equipment to work within manufacturer's recommended safe limits of rated duty;
- b) See that no overload operations are permitted on any installations;
- c) All the cables, switchgears, motors and other ancillary apparatus are of capacity 20% higher than normal operating needs;
- d) All overloads, earth leakage and other protective relays are of correct size regularly tested, maintained and monitored.
- e) All the cables are properly laid and the cable joints are properly supported.
- f) Reducing the number of plug and socket electrical connections; for example, cable couplers;
- g) Avoiding hot surfaces and frictional sparking through correct installation, commissioning and by regular inspection, testing and maintenance, including periodic and effective monitoring of electrical installations, mechanical installations, switch gears, motors, bearings, brake units etc;
- h) Checking that moving parts are not unintentionally rubbing against fixed objects;
- i) Ensuring proper lubrication;
- j) Closely monitoring compound filled couplers,
- k) Using only fire resistant hydraulic fluid in all equipment in underground. (Ref: DGMS Circular No. 10/ 77 and 3/2003).
- l) Prohibiting the use of flame cutting unless hot permit is given and safety precautions are in place;

13.1.4 Fuel & Potential Ignition Sources are kept apart by

- a) Using flame proof and intrinsically safe electrical equipment's.
- b) Separating electric cables, fuel and hydraulic hoses and pipes from hot components by routing and the provision of physical barriers;
- c) Storing flammable materials away from electrical sub-stations and other potential ignition sources.
- d) Training operating personnel in fire fighting so that at least two persons in each shift are available at all locations of firefighting equipment.

13.1.5 Code adopted for reducing conveyor fires:

- a) Using fire resistant conveyor belting;

- b) Ensuring that the conveyor is of sufficient capacity to carry the maximum expected load;
- c) Maintain conveyor system to ensure maximum specified loads are not exceeded on rollers and idlers;
- d) Designing and constructing transfer and loading points so as to minimise spillage and dust.
- e) Ensuring that all parts of the conveying system are accessible for inspection and can be safely cleaned;
- f) Using fire resistant materials in roadways at transfer and loading points;

13.1.6 Code adopted for reducing fire due to Welding & Gas Cutting operations: Special attention is drawn to DGMS circulars No.55 of 1966 on welding belowground and Tech cir no.12 of 2002. To prevent flash back burn to welders. Additionally a detailed safe operating procedure has been formed and circulated to all concerned personnel

SAFE OPERATING PROCEDURE FOR GAS/ARC, CUTTING/ WELDING

01. CUTTING and WELDING procedures should be established which shall minimize the risk of a mine fire. This shall cover the provisions of Reg 118 (2) of CMR 1957 and DGMS Cir. No. 55 of 1966.
02. Before using welding and gas cutting equipment, a careful assessment of the risks should be carried out. This is particularly important if the work is being carried out in unfamiliar surroundings.
03. The main hazards are from fire and explosion. These are caused by:
 - a) Careless handling of a lighted blowpipe resulting in burns to the user or others;
 - b) Using the blowpipe too close to combustible material;
 - c) Cutting up or repairing tanks or drums which contain or may have contained flammable materials;
 - d) Gas leaking from hoses, valves and other equipment;
 - e) Misuse of oxygen;
 - f) Backfires and flashbacks.
 - g) Electrical shock and electrical burns due to leakage of electricity or open sparks or flashes at the point of welding or cables or earthing contact.
04. This operation should be conducted by persons trained to recognize the fire hazards associated with cutting and welding. The training shall include:-
 - a) The safe use of the equipment;
 - b) The precautions to be taken;
 - c) The use of fire extinguishers and

- d) The means of escape, raising the fire alarm and calling the fire
- e) brigade.

05. Manager shall authorise in writing a responsible person (not below the rank of Overman/ Mining Sirdar/Foreman) for each cutting and welding operation.

06. Manager shall authorise welders in writing for cutting and welding.

07. The welder shall check his equipment of welding / cutting operations for effective functioning of regulators, pressure gauges hoses and flame arresters. The hoses should not contain any leaks. The following precautions will prevent leakage –

- a) Keep hoses clear of sharp edges and abrasive surfaces or where vehicles can run over them;
- b) Do not allow hot metal or spatter to fall on hoses;
- c) Handle cylinders carefully. Keep them in an upright position and fasten them to prevent from falling or being knocked over. For example, chain them in a wheeled trolley or against a wall;
- d) Always turn the gas supply off at the cylinder when the job is finished;
- e) Maintain all equipment and keep in good condition;
- f) Regularly check all connections and equipment for faults and leaks.

08. Use the correct gas pressures and nozzle size for the job. In particular, the acetylene pressure must not exceed 0.62 bar (9 psi);

09. Safety measures: Ensure flashback arresters onto the regulator, on both the acetylene and oxygen supply to protect the cylinders. Arresters may be fitted on the blowpipe but these do not give protection from a fire starting in the hose. For long lengths of hose, you should fit arresters on both the blowpipe and the regulator. The fitting of a flashback arrester should not be considered as a substitute for safe working practice.

10. The welding cable will be maintained in good order and properly insulated.

11. All cable joints will be properly made and insulated. The welding cable shall not be allowed to be adjacent to other cables or lie over other cables while welding.

12. The earth lead will be continuous and firmly earthed (ground) to avoid generation of any open sparks.

13. A record of all welding operations will be maintained by concerned Engineer or Pit Engineer and countersigned by the Asst. Manager in charge. The record shall

contain the details like place of welding, nature of the job, names of the competent persons authorised, time and duration of welding operations, and additional precautions other than the safe operating procedure if any.

14. Provide good ventilation at the site of welding. Small leaks may not be detected immediately. If they leak over a period of time into a poorly ventilated room or confined space, a dangerous concentration of gas may accumulate. To prevent gas accumulating:

- a. always provide adequate ventilation during welding and cutting operations;
- b. store gas cylinders outside whenever possible or in a well-ventilated place;
- c. avoid taking gas cylinders into poorly ventilated rooms or confined spaces.

15. All inflammable material will be removed from the site like grease, oil, coal dust, etc., before commencing the work.

16. All the combustible material in the vicinity will be drenched with water or completely covered with a layer of sand or stone dust.

17. Check inflammable gas percentage with methanometer at frequent intervals before and during the welding / cutting operations. The work will be **discontinued** if any inflammable gas is detected.

18. Provide suitable non-inflammable tray beneath the welding site, to catch all sparks and hot particles produced during welding or cutting operations. The tray shall consist of sheet metal covered with wet sand.

19. Wear appropriate personal protective clothing and equipment to prevent personal injury. Preventing injury- The following precautions will help to prevent injury:

- a. Work in a safe location away from other people;
- b. Wear protective clothing, gauntlets and eye protection;
- c. Shut off the blowpipe when not in use. Do not leave a lighted
- d. Blow pipe on a bench or the floor as the force of the flame may cause it to move;
- e. Clamp the work-piece, do not hold it by hand;
- f. Keep hoses away from the working area to prevent contact with flames, heat, sparks or hot spatter.

20. Do not apply heat to an enclosed vessel or confined spaces such as fuel tanks, chemical tanks, and lubricant or Oil drums containing inflammable and combustible materials, mounted tyres, or mechanical equipment with enclosed space. Drums and Tanks containing fuel residue- People have been killed and seriously injured by such explosions. Tanks and drums that are 'empty' usually still have residues in the bottom, and in seams and crevices. Just a teaspoon of flammable liquid in a drum can be enough to cause an explosion when heated and turned into vapour. You must never use an oxy/fuel gas blowpipe on a drum or tank that has contained or may have contained flammable material unless you know it has been made safe.
21. At least two portable fire extinguishers, a bucket with water, and adequate sand or stone dust will be kept in readiness at the site.
22. All the electrical apparatus will be screened from the welding site, by a metal sheet of adequate size.
23. The responsible persons shall examine the place of welding and all surroundings soon after welding operations are completed and keep a watch for further a period of two hours for any signs of heating.
24. The results of the inspection will be recorded in a bound pagged book kept for the purpose.

Responsibility: Welders, Foremen, Sirdar, Overman and Section Incharge

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13.1.7 Fire Prevention Plan

The plan has taken into account:

- a) Likely sources of fire, taking into account the presence of fuel, ignition sources and oxygen.
- b) Precautions to be taken to protect against, to detect and to combat the outbreak and spread of fire, avoid or control sources of ignition;
- c) Minimise the amount of flammable materials below ground;
- d) Detect, warn and fight fires.

13.1.8 All equipment used belowground is approved by DGMS and is fire resistant. All personnel have been cautioned that friction, often associated with defective

bearings, particularly on conveyor systems, is the cause of many fires in mines. When bearings fail or start to fail, temperatures can rapidly increase to the ignition point of lubricants and the melting point of plastic and rubber seals, hastening bearing failure, and causing the temperature to rise rapidly. The fluid used in all hydraulic systems at GDK 10A Incline is of approved type. GDK 10A Incline also reduced the likelihood of equipment failure by ensuring that operators use it only within its rated duty. To prevent electrical equipment becoming an ignition source, it has been ensured that the equipment is-

- a) Rated for normal operation and for foreseeable faults;
- b) Sufficiently robust for the harsh operating conditions found below ground;
- c) Protected against mechanical damage – all cables are well protected;
- d) Protected electrically against dangers arising from overloading, leakage current and short circuits;
- e) Maintained and operated in accordance with the manufacturer or supplier's instructions.
- f) Standing orders for fire fighting have been framed, explained to all personnel. Fire extinguishers are provided at all strategic points.

1.9 Additional precautions:

- a) Provide smoke detectors and alarms at every 300 m distance all along the belt conveyor roads.
- b) Do not take grease and oils to below ground more than one day's consumption.
- c) Use only flame proof equipment for all underground electrical equipment.

13.2 SPONTANEOUS HEATING:

13.2.1 Risk Assessment for Spontaneous heating: [Guidelines for spontaneous heating on the basis of crossing point temperature and moisture content (Circular No. Tech.3/94 of DGMS). Crossing point when it is more than 160°C and moisture content is less than 2% the coal is poorly susceptible to spontaneous heating. When crossing point temperature 140-160°C and moisture content is between 2-5% the coal is moderately susceptible to spontaneous heating. When crossing point temperature less than 120-140°C and moisture content is more than 5% the coal is highly susceptible to spontaneous heating]. Crossing point temperature of our coal (No.1 Seam) is 131°C.

13.2.3 Action plan:

- a) The isolation stoppings of all goaved out panels will be continuously monitored and maintained in good condition to ensure that the atmosphere behind the stoppings remains inert and percentage of oxygen is below 5%.
- b) If, the % of Oxygen reaches more than 5%, then immediately Nitrogen will be flushed in to the sealed off area,
- c) In case, still there exists danger of fire, pressure balancing will be done across the sealed off area to prevent breathing of air into the panel.

13.2.5 Risk Assessment of spontaneous heating due to Subsidence Area: The maximum subsidence predicted was about 2 m over one of the depillaring panels. The subsidence cracks are filled by dozing and dumping Overburden from near by Opencast mines to prevent fresh air entering in to the goaved out area and threat of fire. The area is regularly inspected to see whether any fresh cracks have developed. In case of any new cracks being observed, the same are being immediately filled up by suitably dozing fresh earth.

13.2.6 Training in fire fighting:

All the under ground machine operators, mechanics and electricians and supervisors are trained and are capable of using the fire fighting equipment available at different sites to quench any ignition of electrical cables, switch gears and any other equipment. Fire fighting equipment stipulated in Regulation 120 of CMR 1957 has been installed, and maintained for use in case of any need.

13.3 COAL DUST AND NOISE HAZARD:

Due to high mechanization in this mine, workers are exposed to noise and dust at a higher level than in conventional mines.

13.3.1 Coal Dust Hazard:

A lot of fine coal dust is generated at this mine because of face mechanization and coal transport by a series of belt conveyors. Lot of this coal dust becomes air borne if adequate dust suppression measures are not ensured. Health hazards associated with respirable dust are well known.

Precautions:

- a) Regular cleaning of spillage coal and accumulated dust along the belt conveyor especially at transfer points will be ensured.
- b) Ensure the treatment of all the galleries with stone dust as per the dusting plan.
- c) Overlap auxiliary ventilation will be adopted at road header face in addition to wet cutting to minimise the dust.
- d) Regular sprinklers or manual spraying will be adopted at all coal transfer points.
- e) Frequent water spraying will be ensured all along the belt conveyor to keep the coal moist to reduce air borne dust.
- f) Dust monitoring instruments will be procured at the earliest.
- g) Dust respirators of best quality will be procured and used
- h) All the persons exposed to dust will be examined for lung function test at yearly intervals.

13.3.2 Noise Hazard: The level of noise in the road header face and at shearer in longwall face is always to the tune of 90 db (A). But for shorter intervals especially when the shearer reaches the main gate at the end of the cutting cycle, noise level reaches up to 100 db (A). The shearer operator and the longwall face crew are exposed to high noise levels for about 10-15 minutes in his shift time.

Action Plan:

- a) Regular replacement of worn out parts
- b) Use of ear muffs

- c) Audiometry will be conducted on all the persons working in the mechanized faces at yearly intervals.

13.4 INUNDATION:

13.4.1 Inundation threat from surface water:

Seasonal Nallahs: seasonal nallahs (Tumni Nallah) is flowing almost in the western corner of the property from north to south. Three numbers of seasonal nallahs are identified over the property of Jagnathpur B, which collects rainwater from the catchment area, garland drains over the goaved out panels. All the nallahs ultimately feed the water to Ajay River. Hence, it is essential to keep all the nallahs free from obstruction for smooth flow of water and to prevent any rise of water level in the nallahs and Ajay River.

13.4.2 Surface bore holes: All the bore holes drilled for exploration will be plugged with cement concrete to prevent entry of water through them to underground workings.

13.4.3 Subsidence cracks: The maximum subsidence predicted was about 2 m over depillaring panels. The general practice is to fill the subsidence cracks by dozing and dumping Overburden from near by Opencast mine. Garland drains and central trench over the goaved out area have been made to smoothly divert rain water to the nallahs and finally to the Ajay River. These measures prevent threat of any rain water and HFL water from Ajay River entering in to the mine workings.

13.4.6 Monsoon preparation: Manager will form a team of officers and prepares a check list for guidance of the team members by 30th April every year for checking all possible dangers likely to arise due to monsoon. The team completes its work before 15th May and submits their report. Action indicated by the team members is completed before 31st May.

13.4.7 Inundation threat from underground water: The water management by multiple sumps and pumps is cumbersome and complex. A lot of manpower and attention of Engineer and others is spent for maintaining a number of pumps in different locations.. It is planned, only three/ four strategic locations shall be selected

to install high capacity pumps and eliminate multiple sumps and operation of too many pumps.

13.4.8 Goaved out panels: it is estimated that maximum quantity of water will flow in to the workings from goaved out panels. The make of water in the development headings is almost negligible. All the make of water from the goaved out and working panels reaches different sumps. Adequate capacity pumps installed in multiple sumps pump out the water to different upper level sumps and ultimately to surface.

Analysis: In case of failure of pumping for a considerable time there is a rare chance of water accumulating and drowning abandoned workings. However the following action shall be taken to prevent any danger from this source.

Action plan:

- a) Monitor water level at all the sumps,
- b) Inform Manager, Jagnathpur B when the water level reaches a danger level
- c) Inform all the Supervisory staff of Jagnathpur B to keep a watch for unusual seepage of water
- d) Supervisory staff / workers shall immediately inform Manager or any senior officer present in the mine about unusual seepage of water
- e) Manager / Senior Officer shall take decision for withdrawal of persons.

Responsibility: Manager / senior officer.

13.4.9. the most dangerous situation can arise either because of breakdown of pumps and / or power failure combined with unusually heavy rain fall and sudden inflow of water from catchment area in to the workings. There is no chance of sudden inundation of underground workings in such an event. Management is likely to get adequate time to withdraw all persons from both underground workings of Jaganathpur B. However, water entering and flooding the mines will cause huge loss to the company and normal operation of coal production will be severely effected. Hence, we must take foolproof precautions to rule out the possibility of surface water entering underground workings.

The following precautions will be taken:

- 1. Alternative power arrangements, will be provided
- 2. A Diesel Generator capable of running pumps will be provided

3. All the drains, leading to the mouth of inclines, shafts will be kept free from silt and cleaned before arrival of monsoon
4. The entry points like incline mouth and shafts will be constructed at 3m higher level than the HFL at that point

13.9 Disaster Management Plan: A copy of disaster management plan is kept available at office of the Manager, Jagnathpur B. The evacuation procedure and emergency organization including duties of various officials is indicated in this document to deal with the situation of fire, inundation and major roof fall.

13.9.1 Managing an emergency due to occurrence of FIRE

Immediate steps to be taken:

- a) On observing fire or smoke, evacuate all the persons from the districts except those necessary to deal with the fire.
- b) Immediately communicate to man way clerk, in turn to inform the Manager or Senior most Officers available.
- c) Ensure all the persons use escape route. It is the responsibility of Sirdar, Overman and Assistant Manager.
- d) Pool up the available rescue persons with in the mine or district It is the responsibility of Overman and Assistant Manager.
- e) Intimate the position to the immediate Officer In-charge of the shift. Then shift In-charges should decide whether persons from other districts to be withdrawn or not, which will depend upon the extent of the fire and its likely effect on other districts.

Guidelines to the shift In-charge:

- f) If the fire is small and detected in the incipient stage, it can be quenched with available fire extinguishers of the district. (Example: Fire in respect of switchgear, hauler motor or drill panel etc.,)
- g) If the fire cannot be quenched by the available equipment, it will be safer to assume greater danger to other districts and steps shall be taken to withdraw all the persons from other districts also. (Ex: Pillar fire, goaf fire etc. which needs complete isolation and takes longer time for controlling the Fire).

- h) Shift Incharge should communicate the latest status of the fire to the Manager through Man way clerk.

Manager in turn will initiate action

Subsequent steps:

- i) Duties of Manager: Immediately he should proceed to the mine after receiving the information of fire. He should analyse the available information and inform the Agent, GM and Rescue station. Proceed to Underground and take necessary steps for further evacuation of left over persons. He should give clear guidelines as to the path of travel for escape of trapped persons based on the situation of the fire and presence of toxic gases.
- j) Manager should inspect the site to assess the respirable and irrespirable zone, with the help of safety lamp, CO-detector, Methonometer. He should also decide most suitable place for (FAB) Fresh Air Base to deal the Fire. (Respirable air means oxygen is more than 17% and CO is less than 0.005%).
- k) Arrange for procurement, transportation and stocking of materials like brattice cloth, hallow blocks, stone, cement, sand, props etc., up to fresh air base.
- l) He should make arrangements to pool essential manpower like rescue trained persons, masons, support men etc.
- m) He should instruct Surveyor for preparation of rescue and recovery plans. Extra copies of these plans shall be prepared and should be made available at FAB. (These plans are to be carried by the rescue persons whenever required).
- n) The FAB should have basic facilities like lighting, communication, drinking water supply and catering arrangements. A doctor also should be placed along with minimum supporting staff and medical equipment.
- o) The area in and around the occurrence of the fire must be treated with adequate quantities of dry stone dust. This dusting operation should however to be carried out without causing the slightest hindrance to the main rescue and recovery operation. This dusting helps in preventing coal dust explosion.

- p) Strict security should be posted not to allow any person into the return airway. The persons going into the mine and coming out of the mine shall be clearly recorded. All the relevant records like time, purpose etc., to be incorporated.
- q) Reversal of the main mechanical ventilator is a major decision and should be decided at the level of DMC/CC. Even if it is done it should be done only for limited period for withdrawal of persons in the mine. The pros and cons of the effects of reversal of the fan should be critically analysed.
- r) Temporary arrangements for analytical lab should be established on the mine surface to have faster air samples analysis.
- s) All the rescue and recovery work by the rescue teams wearing breathing apparatus must comply with statutory provisions of Mines Rules.
- t) The status of rescue work should be assessed at every half an hour intervals and communicated to surface control room and in turn to DMC (Disaster Management Committee). This report should cover the progress of work, the need for any additional resources or any new developments in the existing situation.
- u) Persons who saved themselves by barricading from irrespirable zone should be rescued by team of rescue persons with the help of oxygen self-rescuers. If air contains enough oxygen, chemical filter type rescuers can also be used for bringing out these persons.
- v) For persons who are affected by CO poisoning, oxygen resuscitators should be applied and the person has to be carried on a stretcher. All persons rescued shall report to the doctor at FAB and further course of action should be decided on his advice.
- w) Top priority should be given for recovery of survived / trapped persons. Next important priority should be given for recovery of dead bodies.

13.9.2 MANAGING AN INUNDATION DISASTER

- a) At the first indication of unusual seepage, rising water level or threat of inundation, withdraw all the persons from the mine in the areas likely to be effected.
- b) The withdrawal of workmen from the dip most point of the mine is of the first priority. This is the responsibility of shift in-charge Sirdar, Overman and Assistant Manager.
- c) Workers themselves try to run away from the drowning work areas towards rise. In case the outlet from the mine is cut off from the rising water level, there is a possibility of persons taking refuge in the rise side workings (air pockets).
- d) The Sirdar or Overman of the district who has seen the rising water should intimate to the shift In-charge and simultaneously to Manager of the mine through manway clerk / Telephone operator.
- e) The shift incharge should immediately inspect the affected area and assess the situation and give his first hand information to his superiors and also to the Manager. The Manager should summon the rescue persons and doctors. He should also intimate the Agent, Nodal Officers and General Manager.
- f) The rescue of such persons who took shelter in air pockets is done by locating the likely air pockets from the study of the mine plans. From the surface large diameter holes should be drilled to reach such pockets. Metal capsules are lowered by the help of temporary winches and then persons are rescued one by one from such air pockets.
- g) The Manager should immediately proceed underground and assess the situations as to the extent of the area drowned, availability of power, and ventilation for commissioning of dewatering operations.
- h) If the rising water level has disrupted the ventilation circuit, the same should be established by breaking of ventilation stoppings at the dip most possible point, to connect intake and return air ways for the flow of air current.

- i) The Colliery Manager should assess the number of persons, who have escaped from the mine and who are still taking shelter in any possible air pockets. This estimate will give the number of lives lost during inundation. To decide the number of missing persons the Manager should further check with shift In-charge, Manway clerk, lamp room In-charge and co-workers who escaped.
- j) The Manager should assess the total quantity of water with the help of Surveyor, requirements of pumps and other materials for dewatering of the area.
- k) He should send immediately the detailed report regarding the procurement and supply of pumps, pipelines, men and other equipment required to deal with the situation.
- l) In the mean time the Area General Manager forms the Core-Committee and inspects the site and confirms the assessment made by the Manager with modification if any.
- m) He will coordinate all the activities and oversee smooth functioning of all the Committees.
- n) Engineering Department should start installing the additional pumps available with them and also should procure pumps for the purpose of emergency preparedness stationed at Mines Rescue Station.
- o) Simultaneously telephone and good lighting arrangements should be established at site of operation.
- p) The discharge from the additional pumps installed to dewater the area has to be decided by the DMC/CC depending on the situation like availability of Main sump capacities, total head required to be dealt by the pumps, availability of old workings, necessity of recovery of survivors or casualties.
- q) As the dewatering operations are in progress arrangements shall be made for shifting of pumps and extension of pipelines shall be taken by the Engineer In-charge of the operation.

13.9.3 Managing a Major Roof Fall Disaster

Measures to be observed in case of major roof fall:

Immediate steps:

- a) After an occurrence of any major roof fall, persons should be withdrawn from the working places of the district and intimate the Manager through Manway clerk / Telephone operator.
- b) Immediate inspection of all the working places by the shift Assistant Manager .
- c) Saving of trapped persons is most important. It should be done with the help of the available manpower at the site or from other parts of the mine. The work should be started by Shift Assistant Manager / Overman.
- d) In case of difficulty in rescuing the victims of roof falls with the resources available at the mine, rescue station shall be informed immediately and all available resources including Search Camera, and pneumatic jacks etc., from the rescue station shall be summoned and utilized.

Subsequent steps:

- e) Soon after inspection by the senior officers, the rescue work of trapped persons should be placed under the charge of any senior mining officer
- f) Sufficient care should be taken to avoid further casualties. The help of rescue trained persons available within the district and within the mine should be taken in rescue and recovery operations.
- g) First- hand information should be communicated to the surface after assessing the actual situation at an interval of every half an hour.
- h) If the trapped persons are unable to be rescued immediately, they should be given immediate requirements like air, water and liquid diet with the help of small dia pipes.
- i) If the spot of the victim in the roof fall cannot be located by physical senses (seeing and hearing), Search Camera can be used.
- j) Information obtained by search camera shall also be useful to decide the line of progress to rescue the persons.
- k) If the trapped persons are beneath huge boulders, the same can be lifted by pneumatic lifting bags, hydraulic lifting cylinders and other suitable equipment which are available with the Regional Rescue Stations.

- l) Have a constant dialogue with the trapped persons, keep them in good humour and keep up their morale.
- m) At no point of time one should sacrifice his own safety for rescue and recovery of trapped persons. The safety of persons engaged in rescue work is of paramount importance.
- n) A safe and secure place nearest to the location of the fall must be located and used as base for conducting the work of rescue and recovery of trapped persons. The base shall be under the control of senior officer. Any person or group of persons going beyond this point needs a special permission.
- o) A good amount of care and caution needs to be exercised in dressing, supporting, and debris clearing with an idea to reach the location of trapped persons.
- p) A secondary team should start supporting all vulnerable points surrounding the major fall by additional supports to protect the persons engaged in rescue work.
- q) Presence of more persons at the actual rescue spots leads to confusion and delays rescue work. A permit from the incharge of the base shall be issued for all the persons going beyond the base.
- r) Good lighting arrangements should be provided at the base of rescue operations and up to the point of fall.
- s) A telephone communication should be established from the base to the surface.
- t) If ventilation is sluggish at the place of fall, mobilise additional resources to send more quantity of air.
- u) A standby team of persons should be kept ready at the base to replace or change a team of persons engaged in rescue work as soon as they are tired.
- v) Soon after rescuing the victims from underneath the fall, they shall be administered first-aid treatment in underground and sent to surface for further treatment.

1. Resource Optimization / Recycling

- a) **Waste generation** in underground mining is minimal since mining operations are confined to selective extraction of coal. The major waste in the form of overburden (OB) reduced in opencast mining method does not arise in the underground method. Whatever waste is produced in the form of sedimentary rock during crossing of underground faults, if any, is stored / disposed of in pre formed cavities produced during the underground mining process and no surface disposal is required.

- b) **Mining machinery** used are maintained on the principle of modern Smart Life Cycle Management governed by condition monitoring based preventive and predictive maintenance as per best international practice. Major assemblies / sub-assemblies / of machines are re cycled after being repaired / refurbished. Dirty units are kept separately and returned to OEM, and either refurbished or disposed off. This minimizes the need for disposal of worn out machine parts and prolongs machine life cycle significantly. Eco friendly disposing is also done for hazardous parts. Major high horsepower mining machines being electrically operated, emission would be insignificant.
- c) **Energy** Diesel consumption would be nil or minimal since the machines would be operated by electricity. Electrical power consumption would be optimized by ensuring high power factor. **The use of solar power for surface offices and workshops is planned- if possible.**
- d) **Noise and Vibration** Machines used would conform to permitted underground noise limits and vibration standards as recommended by DGMS. The effect of machine noise and vibration would not be felt on the surface of the mine.
Machines complying in underground are DGMS approved, and conforming to international standards of noise and vibration is a pre-requisite for approval.
- e) **Water management** follows a closed loop. Ground water which enters the mine is stored in underground settling sumps and pumped out to the surface where they are treated in water treatment plants / settlement pond and recycled for underground machine cooling and dust suppression. The excess water will be supplied to surface domestic use to local residents, where required.
- f) **No surface coal processing** is envisaged and therefore no waste in the form of tailings, washery rejects, etc. will be produced

2. **Impoundment, damming, culverting, realignment or other changes to the hydrology of neither watercourses nor aquifers?**

- a) Impoundment, damming, culverting, realignment or other changes to the hydrology of watercourses is not envisaged. Surface Water recharge will be done whenever required.
- b) Impoundment, damming, culverting and realignment of aquifers is not envisaged. However, normal underground mining operations can puncture aquifers in localized points of the mine.